

## Exercise 3.2

1. Consider the universal set  $U = \{x | x \text{ is multiple of } 2 \wedge 0 < x \leq 30\}$ ,  $A = \{x | x \text{ is a multiple of } 6\}$  and  $B = \{x : x \text{ is a multiple of } 8\}$

$$U = \{x | x \text{ is multiple of } 2 \wedge 0 < x \leq 30\}$$

$$U = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30\}$$

(i) List all elements of sets  $A$  and  $B$  in tabular form

$$A = \{x | x \text{ is a multiple of } 6\}$$

$$A = \{6, 12, 18, 24, 30\}$$

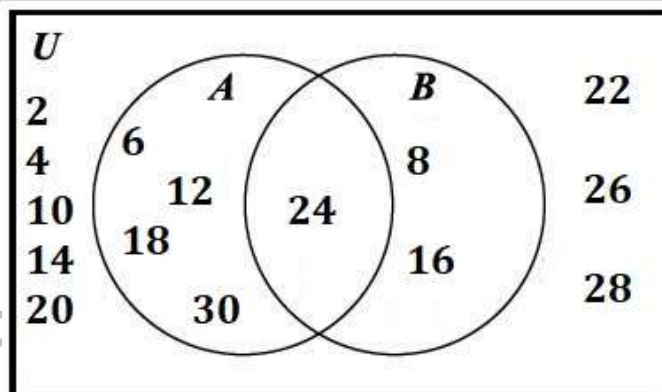
$$B = \{x : x \text{ is a multiple of } 8\}$$

$$B = \{8, 16, 24\}$$

(ii) Find  $A \cap B$

$$\begin{aligned} A \cap B &= \{6, 12, 18, 24, 30\} \cap \{8, 16, 24\} \\ &= \{24\} \end{aligned}$$

(iii) Draw a Venn Diagram



2. Let  $U = \{x | x \text{ is an integer} \wedge 0 < x \leq 150\}$ ,  $G = \{x | x = 2^m \text{ for integer } m \wedge 0 \leq m \leq 12\}$  and  $H = \{x | x \text{ is a square}\}$

$$U = \{x | x \text{ is an integer} \wedge 0 < x \leq 150\}$$

$$U = \{1, 2, 3, \dots, 150\}$$

(i) List all elements of sets  $G$  and  $H$  in tabular form

$$G = \{x | x = 2^m \text{ for integer } m \wedge 0 \leq m \leq 12\}$$

$$G = \{1, 2, 4, 8, 16, 32, 64, 128\}$$

$$H = \{x | x \text{ is a square}\}$$

$$H = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144\}$$

**Note:** For  $G$  Compute  $2^m \leq 150$  and for  $H$  Compute  $k^2 \leq 150$

(ii) Find  $G \cup H$

$$\begin{aligned} G \cup H &= \{1, 2, 4, 8, 16, 32, 64, 128\} \cup \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144\} \\ &= \{1, 2, 4, 8, 9, 16, 25, 32, 36, 49, 64, 81, 100, 121, 128, 144\} \end{aligned}$$

(iii) Find  $G \cap H$

$$\begin{aligned} G \cap H &= \{1, 2, 4, 8, 16, 32, 64, 128\} \cap \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144\} \\ &= \{1, 4, 16, 64\} \end{aligned}$$

3. Consider the set  $P = \{x | x \text{ is a prime number} \wedge 0 < x \leq 20\}$  and  $Q = \{x | x \text{ is a divisor of } 210 \wedge 0 < x \leq 20\}$

$$P = \{x | x \text{ is a prime number} \wedge 0 < x \leq 20\}$$

$$P = \{2, 3, 5, 7, 11, 13, 17, 19\}$$

$$Q = \{x | x \text{ is a divisor of } 210 \wedge 0 < x \leq 20\}$$

$$Q = \{1, 2, 3, 5, 6, 7, 10, 14, 15\}$$

**(i) Find  $P \cap Q$**

$$\begin{aligned} P \cap Q &= \{2, 3, 5, 7, 11, 13, 17, 19\} \cap \{1, 2, 3, 5, 6, 7, 10, 14, 15\} \\ &= \{2, 3, 5, 7\} \end{aligned}$$

**(ii) Find  $P \cup Q$**

$$\begin{aligned} P \cup Q &= \{2, 3, 5, 7, 11, 13, 17, 19\} \cup \{1, 2, 3, 5, 6, 7, 10, 14, 15\} \\ &= \{1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, 17, 19\} \end{aligned}$$

**4. Verify the commutative properties of union and intersection for the following pairs of set.**

**(i)  $A = \{1, 2, 3, 4, 5\}$ ,  $B = \{4, 6, 8, 10\}$**

**Commutative Property of Union:  $A \cup B = B \cup A$**

$$\begin{aligned} L.H.S &= A \cup B \\ &= \{1, 2, 3, 4, 5\} \cup \{4, 6, 8, 10\} \\ &= \{1, 2, 3, 4, 5, 6, 8, 10\} \end{aligned}$$

$$\begin{aligned} R.H.S &= B \cup A \\ &= \{4, 6, 8, 10\} \cup \{1, 2, 3, 4, 5\} \\ &= \{1, 2, 3, 4, 5, 6, 8, 10\} \end{aligned}$$

Hence

$$L.H.S = R.H.S$$

**Commutative Property of Intersection:  $A \cap B = B \cap A$**

$$\begin{aligned} L.H.S &= A \cap B \\ &= \{1, 2, 3, 4, 5\} \cap \{4, 6, 8, 10\} \\ &= \{4\} \end{aligned}$$

$$\begin{aligned} R.H.S &= B \cap A \\ &= \{4, 6, 8, 10\} \cap \{1, 2, 3, 4, 5\} \\ &= \{4\} \end{aligned}$$

Hence

$$L.H.S = R.H.S$$

**(ii)  $N, Z$**

Let  $A = N = \{1, 2, 3, \dots\}$  and  $B = Z = \{0, \pm 1, \pm 2, \pm 3, \dots\}$

**Commutative Property of Union:  $A \cup B = B \cup A$**

$$\begin{aligned} L.H.S &= A \cup B \\ &= \{1, 2, 3, \dots\} \cup \{0, \pm 1, \pm 2, \pm 3, \dots\} \\ &= \{0, \pm 1, \pm 2, \pm 3, \dots\} \end{aligned}$$

$$\begin{aligned} R.H.S &= B \cup A \\ &= \{0, \pm 1, \pm 2, \pm 3, \dots\} \cup \{1, 2, 3, \dots\} \\ &= \{0, \pm 1, \pm 2, \pm 3, \dots\} \end{aligned}$$

Hence

$$L.H.S = R.H.S$$

**Commutative Property of Intersection:  $A \cap B = B \cap A$**

$$\begin{aligned} L.H.S &= A \cap B \\ &= \{1, 2, 3, \dots\} \cap \{0, \pm 1, \pm 2, \pm 3, \dots\} \\ &= \{1, 2, 3, \dots\} \end{aligned}$$

$$\begin{aligned} R.H.S &= B \cap A \\ &= \{0, \pm 1, \pm 2, \pm 3, \dots\} \cap \{1, 2, 3, \dots\} \\ &= \{1, 2, 3, \dots\} \end{aligned}$$

**Alternate:**

Let,  $A = N$  and  $B = Z$

**Union**

$$A \cup B = B \cup A$$

$$N \cup Z = Z \cup N$$

$$Z = Z$$

**Hence Proved**

**Intersection**

$$A \cap B = B \cap A$$

$$N \cap Z = Z \cap N$$

$$N = N$$

**Hence Proved**

$$= \{1, 2, 3, \dots\}$$

Hence

$$L.H.S = R.H.S$$

(iii)  $A = \{x | x \in R \wedge x \geq 0\}, B = R$

**Commutative Property of Union:  $A \cup B = B \cup A$**

$$\begin{aligned} L.H.S &= A \cup B \\ &= \{x | x \in R \wedge x \geq 0\} \cup R \\ &= R \end{aligned}$$

$$\begin{aligned} R.H.S &= B \cup A \\ &= R \cup \{x | x \in R \wedge x \geq 0\} \\ &= R \end{aligned}$$

Hence

$$L.H.S = R.H.S$$

**Commutative Property of Intersection:  $A \cap B = B \cap A$**

$$\begin{aligned} L.H.S &= A \cap B \\ &= \{x | x \in R \wedge x \geq 0\} \cap R \\ &= \{x | x \in R \wedge x \geq 0\} \end{aligned}$$

$$\begin{aligned} R.H.S &= B \cap A \\ &= R \cap \{x | x \in R \wedge x \geq 0\} \\ &= \{x | x \in R \wedge x \geq 0\} \end{aligned}$$

Hence

$$L.H.S = R.H.S$$

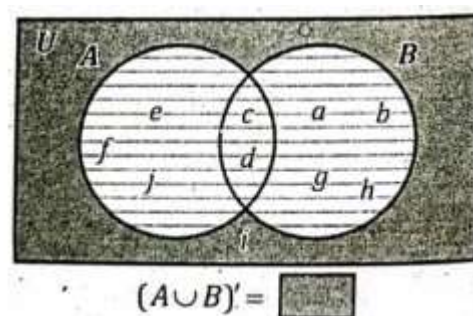
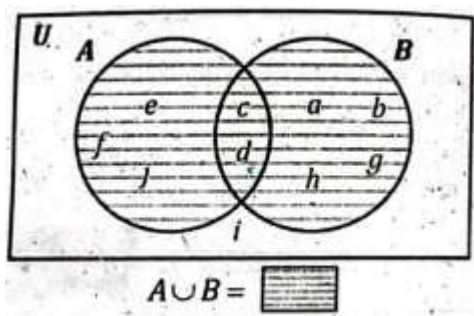
5. Let  $U = \{a, b, c, d, e, f, g, h, i, j\}$ ,  $A = \{a, b, c, d, g, h\}$ ,  $B = \{c, d, e, f, j\}$ , Verify De Morgan's Laws for these sets. Draw Venn Diagram.

**De Morgan's Laws: (i)  $(A \cup B)' = A' \cap B'$  (ii)  $(A \cap B)' = A' \cup B'$**

(i)  $(A \cup B)' = A' \cap B'$

$$\begin{aligned} A \cup B &= \{a, b, c, d, g, h\} \cup \{c, d, e, f, j\} \\ &= \{a, b, c, d, e, f, g, h, j\} \end{aligned}$$

$$\begin{aligned} L.H.S &= (A \cup B)' \\ &= U - (A \cup B) \\ &= \{a, b, c, d, e, f, g, h, i, j\} - \{a, b, c, d, e, f, g, h, j\} \\ L.H.S &= \{i\} \end{aligned}$$



$$\begin{aligned} A' &= U - A \\ A' &= \{a, b, c, d, e, f, g, h, i, j\} - \{a, b, c, d, g, h\} \\ A' &= \{e, f, i, j\} \end{aligned}$$

$$B' = U - B$$

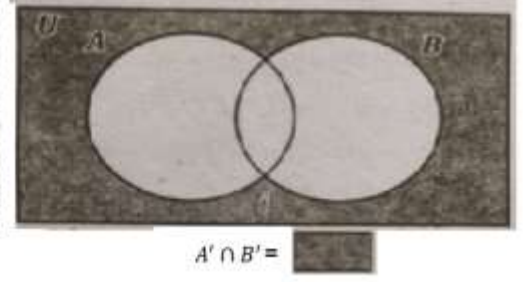
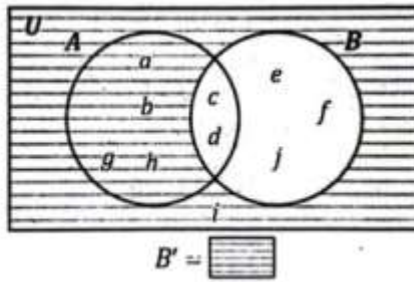
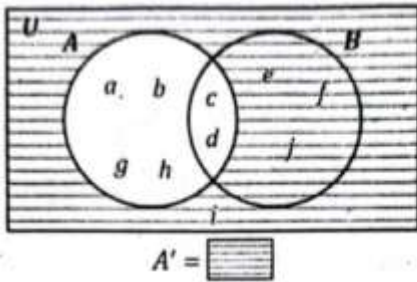
$$B' = \{a, b, c, d, e, f, g, h, i, j\} - \{c, d, e, f, j\}$$

$$B' = \{a, b, g, h, i\}$$

$$R.H.S = A' \cap B'$$

$$= \{e, f, i, j\} \cap \{a, b, g, h, i\}$$

$$R.H.S = \{i\}$$



Hence Proved

$$L.H.S = R.H.S$$

(i)  $(A \cap B)' = A' \cup B'$

$$A \cap B = \{a, b, c, d, g, h\} \cap \{c, d, e, f, j\}$$

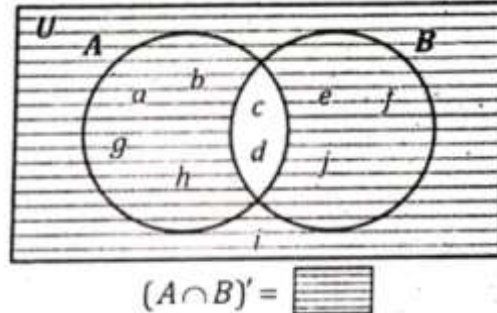
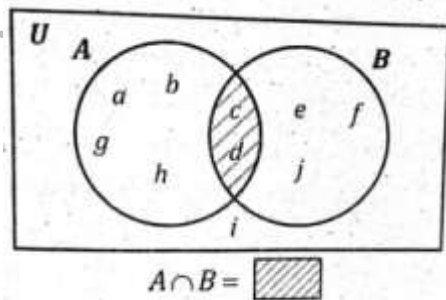
$$= \{c, d\}$$

$$L.H.S = (A \cap B)'$$

$$= U - (A \cap B)$$

$$= \{a, b, c, d, e, f, g, h, i, j\} - \{c, d\}$$

$$L.H.S = \{a, b, e, f, g, h, i, j\}$$



$$A' = U - A$$

$$A' = \{a, b, c, d, e, f, g, h, i, j\} - \{a, b, c, d, g, h\}$$

$$A' = \{e, f, i, j\}$$

$$B' = U - B$$

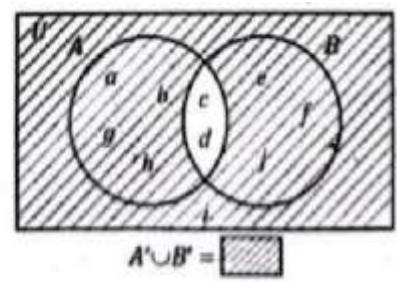
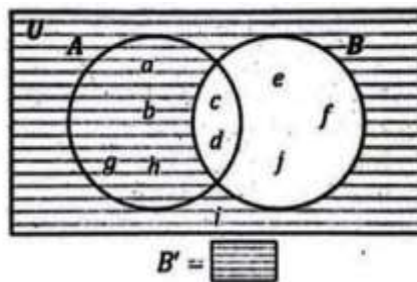
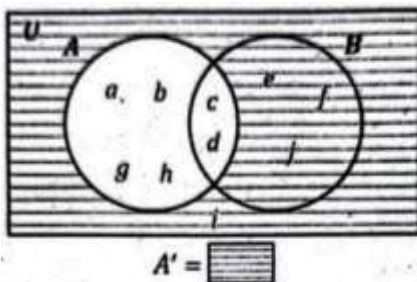
$$B' = \{a, b, c, d, e, f, g, h, i, j\} - \{c, d, e, f, j\}$$

$$B' = \{a, b, g, h, i\}$$

$$R.H.S = A' \cup B'$$

$$= \{e, f, i, j\} \cup \{a, b, g, h, i\}$$

$$R.H.S = \{a, b, e, f, g, h, i, j\}$$



Hence Proved

$$L.H.S = R.H.S$$

6. If  $U = \{1, 2, 3, \dots, 20\}$  and  $A = \{1, 3, 5, \dots, 19\}$ , verify the followings.

(i)  $A \cup A' = U$

$$\begin{aligned}A' &= U - A \\&= \{1, 2, 3, \dots, 20\} - \{1, 3, 5, \dots, 19\} \\A' &= \{2, 4, 6, \dots, 20\} \\L.H.S &= A \cup A' \\&= \{1, 3, 5, \dots, 19\} \cup \{2, 4, 6, \dots, 20\} \\&= \{1, 2, 3, \dots, 20\} \\&= U \\&= R.H.S\end{aligned}$$

(ii)  $A \cap U = A$

$$\begin{aligned}L.H.S &= A \cap U \\&= \{1, 3, 5, \dots, 19\} \cap \{1, 2, 3, \dots, 20\} \\&= \{1, 3, 5, \dots, 19\} \\&= A \\&= R.H.S\end{aligned}$$

(iii)  $A \cap A' = \emptyset$

$$\begin{aligned}A' &= U - A \\&= \{1, 2, 3, \dots, 20\} - \{1, 3, 5, \dots, 19\} \\A' &= \{2, 4, 6, \dots, 20\} \\L.H.S &= A \cap A' \\&= \{1, 3, 5, \dots, 19\} \cap \{2, 4, 6, \dots, 20\} \\&= \{ \} \\&= \emptyset \\&= R.H.S\end{aligned}$$

7. In a class of 55 students, 34 like to play cricket and 30 like to play hockey. Also, each student likes to play at least one of the two games. How many students like to play both games?

Let

$C$  = Set of students who like cricket

$H$  = Set of students who like hockey

According to question:  $n(C \cup H) = 55$ ,  $n(C) = 34$ ,  $n(H) = 30$ ,  $n(C \cap H) = ?$

Using Inclusion-Exclusion Principle,

$$\begin{aligned}n(C \cup H) &= n(C) + n(H) - n(C \cap H) \\55 &= 34 + 30 - n(C \cap H) \\55 &= 64 - n(C \cap H) \\n(C \cap H) &= 64 - 55 \\n(C \cap H) &= 9 \text{ students}\end{aligned}$$

8. In a group of 500 employees, 250 can speak Urdu, 150 can speak English, 50 can speak Punjabi, 40 can speak Urdu and English, 30 can speak both English and Punjabi and 10 can speak Urdu and Punjabi. How many can speak all the three languages?

Let

$U$  = Set of employees who can speak Urdu

$E$  = Set of employees who can speak English

$P$  = Set of employees who can speak Punjabi

According to question:  $n(E \cup U \cup P) = 500$ ,  $n(U) = 250$ ,  $n(E) = 150$ ,  $n(P) = 50$ ,  $n(E \cap U) = 40$ ,  $n(E \cap P) = 30$ ,  $n(U \cap P) = 10$ ,  $n(E \cap U \cap P) = ?$

Using Inclusion-Exclusion Principle,

$$n(E \cup U \cup P) = n(E) + n(U) + n(P) - n(E \cap U) - n(E \cap P) - n(U \cap P) + n(E \cap U \cap P)$$

$$500 = 150 + 250 + 50 - 40 - 30 - 10 + n(E \cap U \cap P)$$

$$500 = 370 + n(E \cap U \cap P)$$

$$500 - 370 = n(E \cap U \cap P)$$

$$130 = n(E \cap U \cap P)$$

$$n(E \cap U \cap P) = \mathbf{130 \text{ employees}}$$

**9. In sports event, 19 people wear blue shirts, 15 people wear green shirts, 3 wear blue and green shirts, 4 wear a cap and blue shirts, 2 wear a cap and green shirts. The total number of people with either a blue or green shirt or a cap is 34. How many people are wearing caps?**

Let

$B = \text{Set of people wearing blue shirts}$

$G = \text{Set of people wearing green shirts}$

$C = \text{Set of people wearing caps}$

According to question:  $n(B) = 19$ ,  $n(G) = 15$ ,  $n(B \cap G) = 3$ ,  $n(C \cap B) = 4$ ,  $n(C \cap G) = 2$ ,  $n(B \cup G \cup C) = 34$ , Assume  $n(B \cap G \cap C) = 0$  (no one wears all three),  $n(C) = ?$

Using Inclusion-Exclusion Principle,

$$n(B \cup G \cup C) = n(B) + n(G) + n(C) - n(B \cap G) - n(B \cap C) - n(G \cap C) + n(B \cap G \cap C)$$

$$34 = 19 + 15 + n(C) - 3 - 4 - 2 + 0$$

$$34 = 25 + n(C)$$

$$34 - 25 = n(C)$$

$$9 = n(C)$$

$$n(C) = \mathbf{9 \text{ people}}$$

**10. In a training session, 17 participants have laptops, 11 have tablets, 9 have laptops and tablets, 6 have laptops and books and 4 have both tablets and books. Eight participants have all three items. The total number of participants with laptops, tablets or books is 35. How many participants have books?**

Let

$L = \text{Set of participants with laptops}$

$T = \text{Set of participants with tablets}$

$B = \text{Set of participants with books}$

According to question:  $n(L) = 17$ ,  $n(T) = 11$ ,  $n(L \cap T) = 9$ ,  $n(L \cap B) = 6$ ,  $n(T \cap B) = 4$ ,  $n(L \cap T \cap B) = 8$ ,  $n(L \cup T \cup B) = 35$ ,  $n(B) = ?$

Using Inclusion-Exclusion Principle,

$$n(L \cup T \cup B) = n(L) + n(T) + n(B) - n(L \cap T) - n(L \cap B) - n(T \cap B) + n(L \cap T \cap B)$$

$$35 = 17 + 11 + n(B) - 9 - 6 - 4 + 8$$

$$35 = 17 + n(B)$$

$$35 - 17 = n(B)$$

$$18 = n(B)$$

$$n(B) = \mathbf{18 \text{ participants}}$$

**11. A shopping mall has 150 employees labelled 1 to 150, representing the Universal set  $U$ . The employees fall into the following categories:**

- Set A: 40 employees with a salary range of 30k - 40k, labelled from 50 to 89.
- Set B: 50 employees with a salary range of 50k - 80k, labelled from 101 to 150.
- Set C: 60 employees with a salary range of 100k - 150k, labelled from 1 to 49 and 90 to 100

**(a) Find  $(A' \cup B') \cap C$**

$$U = \{1, 2, 3, 4, \dots, 150\}$$



$$A = \{50, 51, 52, \dots, 89\}$$

$$B = \{101, 102, 103, \dots, 150\}$$

$$C = \{1, 2, 3, \dots, 49, 90, 91, 92, \dots, 100\}$$

$$A' = U - A$$

$$A' = \{1, 2, 3, 4, \dots, 150\} - \{50, 51, 52, \dots, 89\}$$

$$A' = \{1, 2, \dots, 49, 90, 91, 92, \dots, 150\}$$

$$B' = U - B$$

$$B' = \{1, 2, 3, 4, \dots, 150\} - \{101, 102, 103, \dots, 150\}$$

$$B' = \{1, 2, 3, \dots, 100\}$$

$$A' \cup B' = \{1, 2, \dots, 49, 90, 91, 92, \dots, 150\} \cup \{1, 2, 3, \dots, 100\}$$

$$A' \cup B' = \{1, 2, 3, \dots, 150\}$$

$$(A' \cup B') \cap C = \{1, 2, 3, \dots, 150\} \cap \{1, 2, 3, \dots, 49, 90, 91, 92, \dots, 100\}$$

$$(A' \cup B') \cap C = \{1, 2, 3, \dots, 49, 90, 91, 92, \dots, 100\}$$

**(b) Find  $n\{A \cap (B' \cap C')\}$**

$$B' = U - B$$

$$B' = \{1, 2, 3, 4, \dots, 150\} - \{101, 102, 103, \dots, 150\}$$

$$B' = \{1, 2, 3, \dots, 100\}$$

$$C' = U - C$$

$$C' = \{1, 2, 3, 4, \dots, 150\} - \{1, 2, 3, \dots, 49, 90, 91, 92, \dots, 100\}$$

$$C' = \{50, 51, 52, \dots, 89, 101, 102, 103, \dots, 150\}$$

$$B' \cap C' = \{1, 2, 3, \dots, 100\} \cap \{50, 51, 52, \dots, 89, 101, 102, 103, \dots, 150\}$$

$$B' \cap C' = \{50, 51, 52, \dots, 89\}$$

$$B' \cap C' = A$$

$$n\{A \cap (B' \cap C')\} = n\{A \cap A\}$$

$$= n(A)$$

$$n\{A \cap (B' \cap C')\} = 40$$

**12. In a secondary school with 125 students participate in at least one of the following sports: Cricket, Football and Hockey.**

- 60 students play cricket.
- 70 students play football.
- 40 students play hockey.
- 25 students play both cricket and football.
- 15 students play both football and hockey.
- 10 students play both cricket and hockey.

**(a) How many students play all the three sports?**

Let

$C$  = Set of students who play cricket  
 $F$  = Set of students who play football  
 $H$  = Set of students who play hockey

According to question:  $n(C \cup F \cup H) = 125$ ,  $n(C) = 60$ ,  $n(F) = 70$ ,  $n(H) = 40$ ,  $n(C \cap F) = 25$ ,  $n(F \cap H) = 15$ ,  $n(C \cap H) = 10$ ,  $n(C \cap F \cap H) = ?$

Using Inclusion-Exclusion Principle,

$$n(C \cup F \cup H) = n(C) + n(F) + n(H) - n(C \cap F) - n(F \cap H) - n(C \cap H) + n(C \cap F \cap H)$$

$$125 = 60 + 70 + 40 - 25 - 15 - 10 + n(C \cap F \cap H)$$

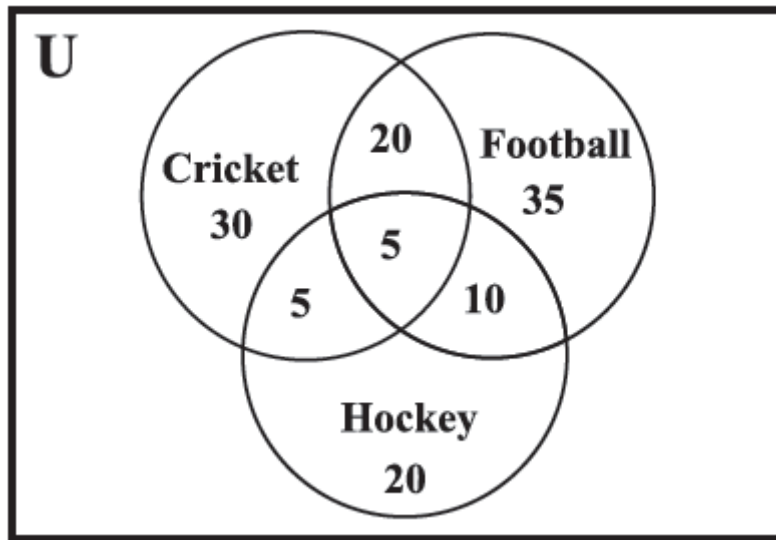
$$125 = 120 + n(C \cap F \cap H)$$

$$125 - 120 = n(C \cap F \cap H)$$

$$5 = n(C \cap F \cap H)$$

$$n(C \cap F \cap H) = 5 \text{ students}$$

(b) Draw a Venn Diagram showing the distribution of sports participation in all the games.



**Note:** The Venn diagram can be constructed as follows:

1. **All three sports (center):** 5 students.
2. **Cricket and Football only:**  $25 - 5 = 20$  students.
3. **Football and Hockey only:**  $15 - 5 = 10$  students.
4. **Cricket and Hockey only:**  $10 - 5 = 5$  students.
5. **Cricket only:**  $60 - (20 + 5 + 5) = 30$  students.
6. **Football only:**  $70 - (20 + 10 + 5) = 35$  students.
7. **Hockey only:**  $40 - (10 + 5 + 5) = 20$  students.

**13.** A survey was conducted in which 130 people were asked about their favorite foods. The survey results showed the following information:

- 40 people said they liked nihari
- 65 people said they liked biryani
- 50 people said they liked korma
- 20 people said they liked nihari and biryani
- 35 people said they liked biryani and korma
- 27 people said they liked nihari and korma
- 12 people said they liked all the three foods: nihari, biryani and korma

(a) At least how many people like nihari, biryani or korma?

Let

$N$  = Set of people liked nihari

$B$  = Set of people liked biryani

$K$  = Set of people liked korma

According to question:  $n(U) = 130$ ,  $n(N) = 40$ ,  $n(B) = 65$ ,  $n(K) = 50$ ,  $n(N \cap B) = 20$ ,  $n(B \cap K) = 35$ ,  $n(N \cap K) = 27$ ,  $n(N \cap B \cap K) = 12$

Using Inclusion-Exclusion Principle,

$$n(N \cup B \cup K) = n(N) + n(B) + n(K) - n(N \cap B) - n(B \cap K) - n(N \cap K) + n(N \cap B \cap K)$$

$$n(N \cup B \cup K) = 40 + 65 + 50 - 20 - 35 - 27 + 12$$



$$n(N \cup B \cup K) = 85 \text{ people}$$

**(b) How many people did not like nihari, biryani or korma?**

$$\begin{aligned} n(N \cup B \cup K)' &= n(U) - n(N \cup B \cup K) \\ &= 130 - 85 \\ n(N \cup B \cup K)' &= 45 \text{ people} \end{aligned}$$

**(c) How many people like only one of the following foods: nihari, biryani or korma?**

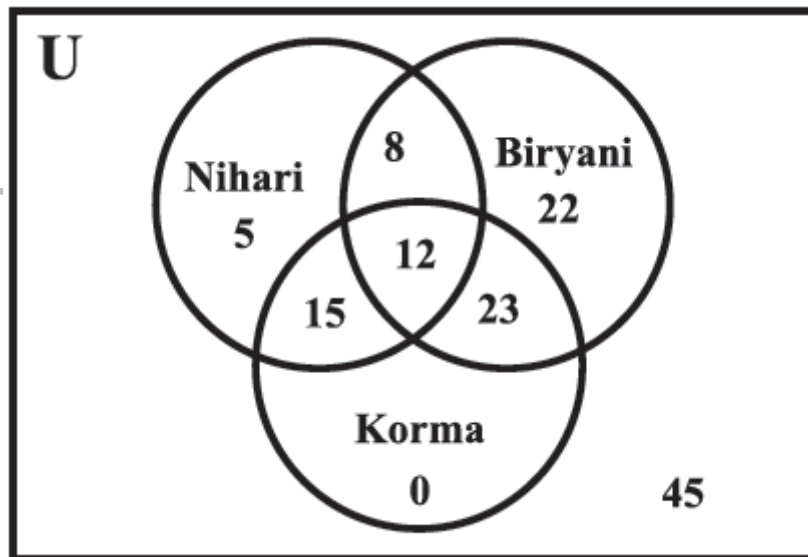
$$\begin{aligned} \text{People like only Nihari} &= n(N) - n(N \cap B) - n(N \cap K) + n(N \cap B \cap K) \\ &= 40 - 20 - 27 + 12 \\ &= 5 \text{ people} \end{aligned}$$

$$\begin{aligned} \text{People like only Biryani} &= n(B) - n(N \cap B) - n(B \cap K) + n(N \cap B \cap K) \\ &= 65 - 20 - 35 + 12 \\ &= 22 \text{ people} \end{aligned}$$

$$\begin{aligned} \text{People like only Korma} &= n(K) - n(B \cap K) - n(N \cap K) + n(N \cap B \cap K) \\ &= 50 - 35 - 27 + 12 \\ &= 0 \end{aligned}$$

$$\text{Total} = 5 + 22 + 0 = 27 \text{ people like only one food}$$

**(d) Draw a Venn Diagram.**



**Note:** The Venn diagram can be constructed as follows:

1. **All three foods (center):** 12 people.
2. **Nihari and Biryani only:**  $20 - 12 = 8$  people.
3. **Biryani and Korma only:**  $35 - 12 = 23$  people.
4. **Nihari and Korma only:**  $27 - 12 = 15$  people.
5. **Only Nihari:**  $40 - (8 + 12 + 15) = 5$  people.
6. **Only Biryani:**  $65 - (8 + 12 + 23) = 22$  people.
7. **Only Korma:**  $50 - (23 + 12 + 15) = 0$  people.